

Haleh Damirchi

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Machine learning researcher and PhD graduate from Queen's University specializing in computer vision, probabilistic forecasting, and pedestrian trajectory prediction.

Work Experience

Graduate Research Fellow

May 2021 – April 2026

Queen's University

AIIM lab, RCV lab

- Co-developed a multitask crowd counting and localization network using pretrained VGG features and point-level supervision.
- Designed and developed context-aware multimodal trajectory forecasting models using Transformer-based and recurrent architectures.
- Designed a socially informed trajectory forecasting framework combining CVAE-based forecasting, trajectory reconstruction, and online pseudo-trajectory augmentation for challenging training samples.
- Conducted theoretical and empirical analysis of evaluation methods for pedestrian trajectory forecasting to improve the rigor of model comparisons.
- Published peer-reviewed research at WACV, ICIP, and ICASSP.

University Lecturer

Jan 2026 – April 2026

Queen's University

Introduction to Data Science

- Delivered weekly lectures and provided responsive academic support to a large undergraduate class.
- Led lab sessions and guided students through assignments, projects, and core data science concepts.
- Developed lecture materials and designed assessments aligned with course learning objectives.

Teaching Assistant

2019 – 2025

Queen's University, Amirkabir University of Technology

- Assisted with Machine Vision with Deep Learning, C Programming, Artificial Intelligence, Machine Learning and Logic Circuits courses.
- Evaluated assignments and exams with careful attention to consistency and academic standards.
- Mentored students through coursework and helped them strengthen debugging and problem solving skills.

R&D Intern

Summer, Fall 2016

Aria Kavosh Industrial Corp.

- Built and wired an educational control unit integrating relays, PLCs, and an HMI interface.
- Developed detailed wiring and logic diagrams in Microsoft Visio to document system design.

R&D Intern

Summer 2014

Tabriz Peguh

- Conducted research on a system for distinguishing between gasoline and diesel fuel in vehicle fuel tanks.

Technical Skills

Programming: Python, C, SQL, MATLAB

ML/AI: PyTorch, TensorFlow, scikit-learn, Transformers, CNNs, RAG, probabilistic forecasting, multimodal learning

Tools: FastAPI, Git, MySQL, L^AT_EX

Publications

- [1] **H. Damirchi**, M. Greenspan and A. Etemad, “A Comprehensive and Critical Analysis of Metrics and Evaluation Practices in Pedestrian Trajectory Forecasting”, Under Review.
- [2] **H. Damirchi**, A. Etemad and M. Greenspan, “Socially-informed Reconstruction for Pedestrian Trajectory Forecasting”, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025). 
- [3] **H. Damirchi**, M. Greenspan and A. Etemad, “Context-Aware Pedestrian Trajectory Prediction with Multimodal Transformer”, International Conference on Image Processing (ICIP 2023). 
- [4] M. Zand, **H. Damirchi**, A. Farley and M. Molahasani, “Multiscale Crowd Counting and Localization by Multitask Point Supervision”, IEEE International Conference on Acoustics (ICASSP 2022). 
- [5] **H. Damirchi**, S. Seyedin and S. M. Ahadi, “Speaker Extraction Using Stacked BLSTM Optimized with Frequency-domain Differentiated Spectrum Loss”, Iranian Conference on Electrical Engineering (ICEE 2020).
- [6] **H. Damirchi**, S. Seyedin, S. M. Ahadi, “Improving the Loss Function Efficiency for Speaker Extraction Using Psychoacoustic Effects”, *Applied Acoustics*.

Education

Queen’s University

PhD. Electrical and Computer Engineering

Kingston, Canada

2021 – 2026

Thesis: Pedestrian Trajectory Forecasting in Urban Scenarios

Amirkabir University of Technology

M.Sc. Electrical Engineering

Tehran, Iran

2017 – 2020

Thesis: Single-channel Speaker Extraction based on Deep Learning

Tabriz University

B.Sc. Electrical Engineering

Tabriz, Iran

2012 – 2016

Selected Projects

SlouchLess | *Python*

- Packaged and published standalone Windows and Linux releases for an application that detects prolonged slouching from webcam video and provides real-time posture reminders. 
- Built a data collection and training pipeline for learning posture patterns.

PaperScraper | *Python*

- Built a retrieval augmented LLM pipeline to extract datasets, sample counts, and evaluation metrics from research papers, with citations to supporting source text. 
- Implemented PDF parsing and OCR, HyDE retrieval with embeddings, and schema-constrained LLM outputs for structured information extraction.
- Developed an automated evaluation pipeline against ground truth using statistical similarity and LLM-as-judge evaluation to measure extraction performance.

Speaker Extraction | *TensorFlow, Pytorch, Matlab*

- Developed stacked BLSTM models for extracting a target speaker from single-channel speech mixtures. 
- Designed frequency-aware and psychoacoustic loss functions using Mel and Gammatone filter banks and perceptual hearing models.

Speech Recognition and Enhancement | *Matlab*

- Used Hidden Markov Models (HMM) and Dynamic Time Warping (DTW) to recognize an utterance.
- Enhanced the input noisy speech signal using MMSE and Spectral Subtraction algorithm.
- Applied Ramirez04 algorithm to detect the activity of speech signals.

Backpropagation using Evolutionary Algorithms | *Python*

- Formulated a deep neural network training approach based on Gray Wolf and Particle Swarm Optimization.

Service and Professional Activities

- **Reviewer:** ICLR, NeurIPS 2026; NeurIPS, AISTATS, ICLR, and ICML 2025; NeurIPS and ICASSP 2024; IEEE Transactions on Artificial Intelligence; IEEE Transactions on Affective Computing
- **Guest Lecturer:** Computer Vision with Deep Learning, Queen's University, Fall 2025
- **Organizing Committee Member:** International Conference on Signal Processing and Intelligent Systems, Amirkabir University of Technology, Dec. 2018

Languages

- Azeri, Persian: Native
- English: Fluent